

DIFFUSION MODELS ARE EVOLUTIONARY ALGORITHMS

Ernest Nasyrov

MIPT

May 11, 2025

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Problem

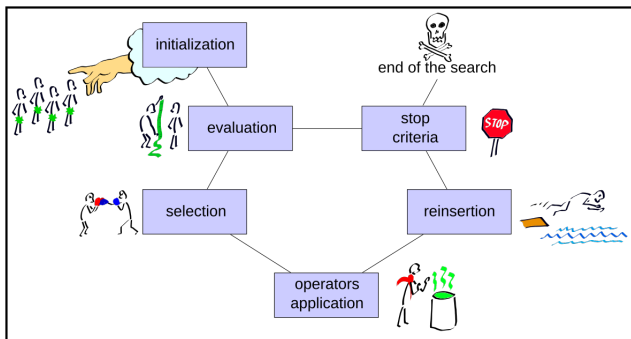
- Develop the mathematical model for an evolution process
- Find the connection between the diffusion models and evolutionary algorithms

The proposal of the paper

Diffusion Evolution Algorithm, which can discover *diverse* solutions, mirroring the diversity of the biosphere.

Evolutionary Algorithms

Main concept: maintain and optimize a population of genotypic parameters over *successive generations* through operations biological operations, such as selection of the fittest, reproduction, genetic crossover, and mutations



General schema of an Evolutionary Algorithm (EA)

Рис.: Evolutionary Algorithms concept

DDPMs

Diffusion process:

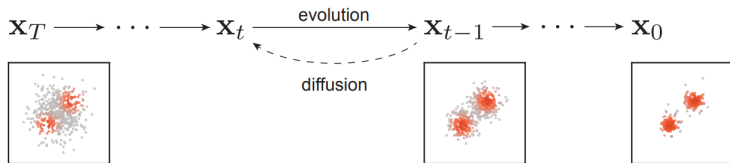
$$\mathbf{x}_t = \sqrt{\hat{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \hat{\alpha}_t} \varepsilon;$$

Loss function:

$$L(\theta) = \mathbb{E}_{\mathbf{x}_0, t, \varepsilon} \|\varepsilon - f_\theta(\mathbf{x}_t, t)\|^2; \quad \hat{\alpha}_t = \prod_{i=1}^t (1 - \beta_i)$$

Sampling:

$$\mathbf{x}_n = \sqrt{\alpha_{t-1}} \hat{\mathbf{x}}_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \hat{\varepsilon} + \sigma_t \omega$$



Diffusion is evolution

Idea: higher *fitness* $\iff$ higher *pdf*.

- ① Fitness function: $f : \mathbb{R}^b \rightarrow \mathbb{R}$
- ② mapping into pdf g : $g[f(x)] = p(x)$. Then $p(x_0 = x) = g[f(x)]$
- ③ In DDPM, the error ε is estimated using neural network θ :

$$\varepsilon = \frac{x_t - \sqrt{\alpha_t}x_0}{\sqrt{1 - \alpha_t}} = \varepsilon_\theta(x_t, t)$$

- ④ In contrast with DDPM, DiffEvo estimates the x_0 from $(x_t, f(x_t))$:

$$p(x_0 = x | x_t) = [\text{Bayes}] = \frac{p(x_t | x_0 = x)g[f(x)]}{p(x_t)}$$

Diffusion is evolution

- 1 In contrast with DDPM, DiffEvo estimates the x_0 from $(x_t, f(x_t))$:

$$p(x_0 = x | x_t) = [\text{Bayes}] = \frac{p(x_t | x_0 = x) g[f(x)]}{p(x_t)}$$

- 2 $p(x_t | x_0 = x) \sim \mathcal{N}(x_t | \sqrt{\alpha_t}x; 1 - \alpha_t)$ — is known
- 3 As our loss is MSE, the estimated x_0 should be the weighted average of the sample x :

$$\hat{x}_0(x_t, \alpha, t) = \sum_{x \sim p_{\text{eval}}(x)} g[f(x)] \frac{\mathcal{N}(x_t | \sqrt{\alpha_t}x; 1 - \alpha_t)}{p(x_t)} \cdot x$$

Idea: by substituting the above formula for the x_0 estimation into diffusion sampling, we get Evolution Diffusion algorithm.

Evolution Diffusion Algorithm example

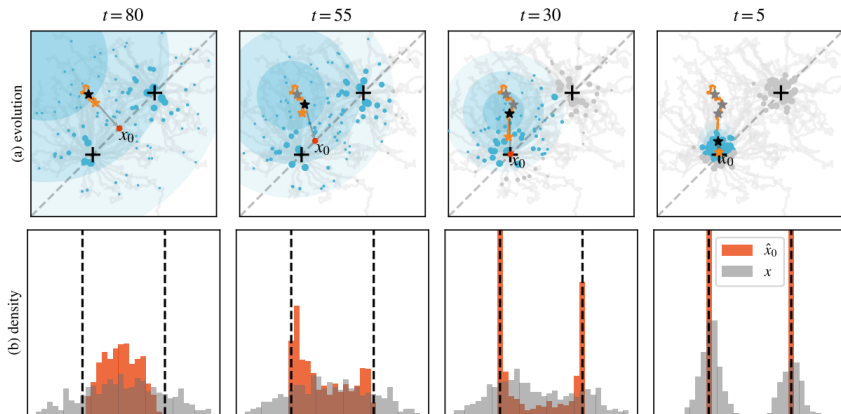


Рис.: DiffEvo algorithm. Individual x_t — black star, $\hat{x}_0$ — red dots, orange star — movement of an individual

<https://arxiv.org/pdf/2410.02543v2>

Experiments

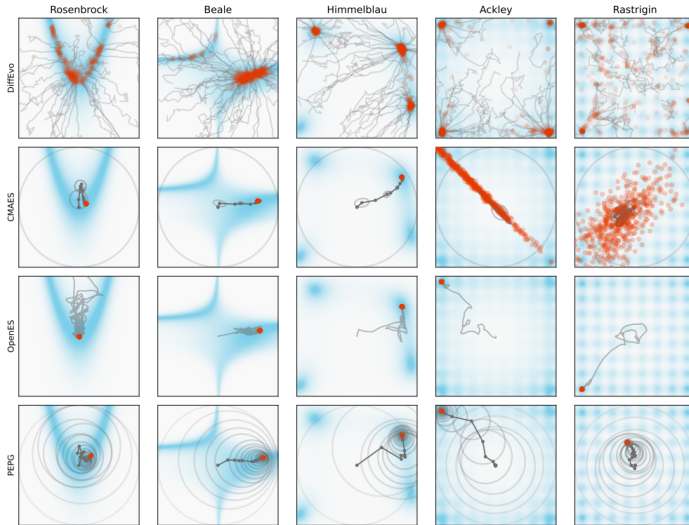


Рис.: Comparison with other algorithms

Pseudocode

Algorithm 1 Diffusion Evolution

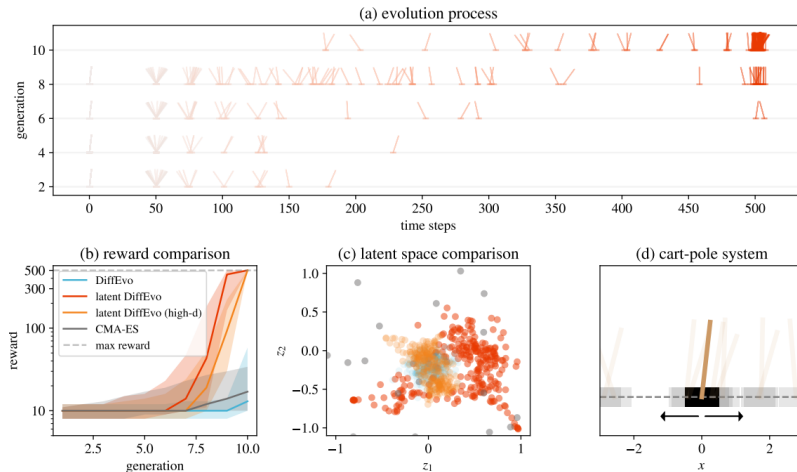
Require: Population size N , parameter dimension D , fitness function f , density mapping function g , total evolution steps T , diffusion schedule α and noise schedule σ .

Ensure: $\alpha_0 \sim 1, \alpha_T \sim 0, \alpha_i > \alpha_{i+1}, 0 < \sigma_i < \sqrt{1 - \alpha_{i-1}}$

```

1:  $[\mathbf{x}_T^{(1)}, \mathbf{x}_T^{(2)}, \dots, \mathbf{x}_T^{(N)}] \leftarrow \mathcal{N}(0, I^{N \times D})$  ▷ Initialize population
2: for  $t \in [T, T-1, \dots, 2]$  do
3:    $\forall i \in [1, N] : Q_i \leftarrow g[f(\mathbf{x}_t^{(i)})]$  ▷ Fitness are cached to avoid repeated evaluations
4:   for  $i \in [1, 2, \dots, N]$  do
5:      $Z \leftarrow \sum_{j=1}^N Q_j \mathcal{N}(\mathbf{x}_t^{(i)}; \sqrt{\alpha_t} \mathbf{x}_t^{(j)}, 1 - \alpha_t)$ 
6:      $\hat{\mathbf{x}}_0 \leftarrow \frac{1}{Z} \sum_{j=1}^N Q_j \mathcal{N}(\mathbf{x}_t^{(i)}; \sqrt{\alpha_t} \mathbf{x}_t^{(j)}, 1 - \alpha_t) \mathbf{x}_t^{(j)}$ 
7:      $\mathbf{w} \leftarrow \mathcal{N}(0, I^D)$ 
8:      $\mathbf{x}_{t-1}^{(i)} \leftarrow \sqrt{\alpha_{t-1}} \hat{\mathbf{x}}_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \frac{\mathbf{x}_t^{(i)} - \sqrt{\alpha_t} \hat{\mathbf{x}}_0}{\sqrt{1 - \alpha_t}} + \sigma_t \mathbf{w}$ 
9:   end for
10: end for
  
```

Latent Space Diffusion Evolution



LSD Pseudocode

Algorithm 2 Latent Space Diffusion Evolution

Require: Population size N , parameter dimension D , latent space dimension d , fitness function f , density mapping function g , total evolution steps T , diffusion schedule α and noise schedule σ .

Ensure: $\alpha_0 \sim 1, \alpha_T \sim 0, \alpha_i > \alpha_{i+1}, 0 < \sigma_i < \sqrt{1 - \alpha_{i-1}}$

```

1:  $\mathbf{E}^{(d,D)} \leftarrow \mathcal{N}(0, 1/D)$  ▷ Initialize the random mapping
2:  $[\mathbf{x}_T^{(1)}, \mathbf{x}_T^{(2)}, \dots, \mathbf{x}_T^{(N)}] \leftarrow \mathcal{N}(0, I^{N \times D})$  ▷ Initialize population
3: for  $t \in [T, T-1, \dots, 2]$  do
4:    $\forall i \in [1, N] : Q_i \leftarrow g[f(\mathbf{x}_t^{(i)})]$  ▷ Fitness are cached to avoid repeated evaluations
5:    $\forall i \in [1, N] : \mathbf{z}_t^{(i)} \leftarrow \mathbf{E} \mathbf{x}_t^{(i)}$  ▷ Encode individual parameters into latent space
6:   for  $i \in [1, 2, \dots, N]$  do
7:      $Z \leftarrow \sum_{j=1}^N Q_j \mathcal{N}(\mathbf{z}_t^{(i)}; \sqrt{\alpha_t} \mathbf{z}_t^{(j)}, 1 - \alpha_t)$ 
8:      $\hat{\mathbf{x}}_0 \leftarrow \frac{1}{Z} \sum_{j=1}^N Q_j \mathcal{N}(\mathbf{z}_t^{(i)}; \sqrt{\alpha_t} \mathbf{z}_t^{(j)}, 1 - \alpha_t) \mathbf{x}_t^{(j)}$ 
9:      $\mathbf{w} \leftarrow \mathcal{N}(0, I^D)$ 
10:     $\mathbf{x}_{t-1}^{(i)} \leftarrow \sqrt{\alpha_{t-1}} \hat{\mathbf{x}}_0 + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \frac{\mathbf{x}_t^{(i)} - \sqrt{\alpha_t} \hat{\mathbf{x}}_0}{\sqrt{1 - \alpha_t}} + \sigma_t \mathbf{w}$ 
11:  end for
12: end for
  
```



Zhang, Yanbo, et al (2024)

Diffusion Models are Evolutionary Algorithms